LAB 21 – Collaborative AI Coding: Pair Programming and Version Control Integration

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BATCH:14

TASK1:

Prompt given to AI:

"Generate a simple Python To-Do List Manager with options to add tasks, view tasks, delete tasks and exit. Then improve the program by adding error handling, input validation and a feature to mark tasks as completed.

Old Code:

```
tasks = [] while

True:

   print("\n===== TO-DO LIST MANAGER =====")

print("1. Add Task")     print("2. View Tasks")

print("3. Delete Task")     print("4. Exit")


   choice = input("Enter your choice: ")


   if choice == "1":
```

```
        task = input("Enter task: ")
        tasks.append(task)
print("Task added successfully!")

    elif choice == "2":         for i, task in
enumerate(tasks, start=1):              print(i,
".", task)

    elif choice == "3":
        task_no = int(input("Enter task number: "))
tasks.pop(task_no - 1)          print("Task deleted
successfully!")

    elif choice == "4":
        break

    else:
        print("Invalid choice!")
```

IMPROVED CODE:

```
tasks = []

while True:
    print("\n===== TO-DO LIST MANAGER =====")
print("1. Add Task")     print("2. View Tasks")
print("3. Delete Task")     print("4. Exit")
```

```
    choice = input("Enter your choice: ")

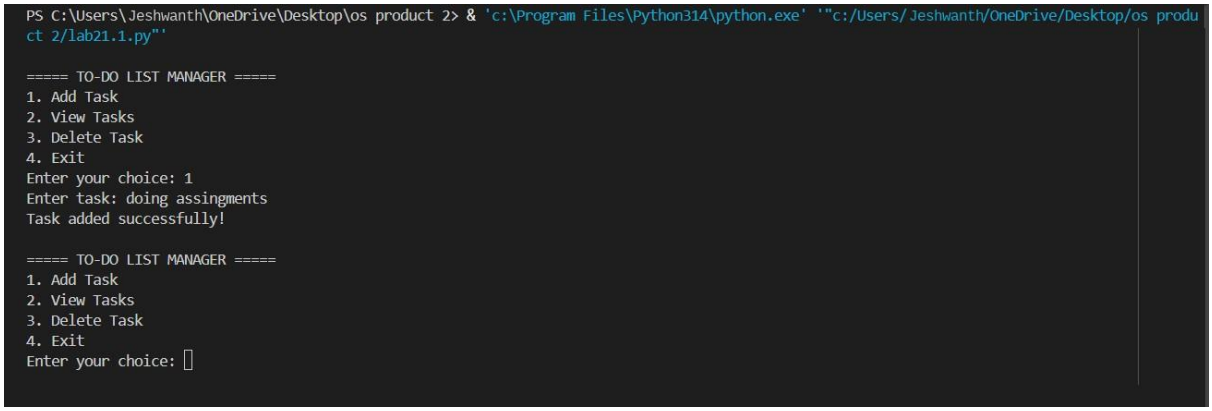
    if choice == "1":        task =
input("Enter task: ")
tasks.append(task)        print("Task
added successfully!")

    elif choice == "2":        for i, task in
enumerate(tasks, start=1):            print(i,
".", task)

    elif choice == "3":
if tasks:
            task_no = int(input("Enter task number: "))
if 1 <= task_no <= len(tasks):
tasks.pop(task_no - 1)               print("Task
deleted successfully!")            else:
print("Invalid task number!")        else:
            print("No tasks to delete!")

    elif choice == "4":
        break
print("Invalid choice!")
```

OUTPUT:



TASK2:

Prompt given to AI:

"Explain how to initialize a Git repository, create and switch to a new branch, add AI-assisted code modifications, commit changes with a meaningful message and merge the changes into the main branch. Provide the Git commands with explanations."

Old Code:

```
tasks = []


while True:
    print("\n===== TO-DO LIST MANAGER =====")
print("1. Add Task")     print("2. View Tasks")
print("3. Delete Task")     print("4. Exit")


    choice = input("Enter your choice: ")


    if choice == "1":         task =
input("Enter task: ")
```

```
tasks.append(task)        print("Task
added successfully!")

   elif choice == "2":        for i, task in
enumerate(tasks, start=1):            print(i,
".", task)

   elif choice == "3":
      task_no = int(input("Enter task number: "))
tasks.pop(task_no - 1)        print("Task deleted
successfully!")

   elif choice == "4":
      break

   else:
      print("Invalid choice!")
```

IMPROVED CODE:

```
tasks = []

def add_task():
   task = input("Enter task: ").strip()

   if task:
      tasks.append({"name": task, "completed": False})
print("Task added successfully!")     else:
```

```
        print("Task cannot be empty!")



def view_tasks():     if not tasks:
print("No tasks available!")
return

   for i, task in enumerate(tasks, start=1):         status =
"Completed" if task["completed"] else "Pending"         print(i,
task["name"], "-", status)



def delete_task():
view_tasks()

   if not tasks:
return

try:
        n = int(input("Enter task number: "))

        if 1 <= n <= len(tasks):
tasks.pop(n - 1)             print("Task
deleted successfully!")         else:
            print("Invalid task number!")
```

```
    except ValueError:
        print("Enter a valid number!")



def complete_task():
view_tasks()

    if not tasks:
return

try:
        n = int(input("Enter task number: "))

        if 1 <= n <= len(tasks):
tasks[n-1]["completed"] = True
print("Task completed!")        else:
            print("Invalid task number!")

    except ValueError:
        print("Enter a valid number!")



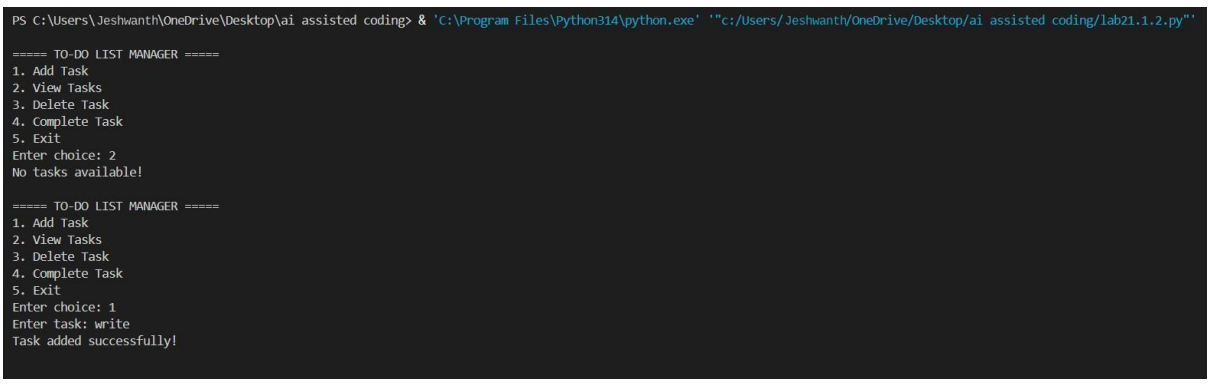
while True:
    print("\n===== TO-DO LIST MANAGER =====")
print("1. Add Task")     print("2. View Tasks")
```

```
print("3. Delete Task")     print("4. Complete
Task")     print("5. Exit")

   choice = input("Enter choice: ")

   if choice == "1":
add_task()     elif
choice == "2":
view_tasks()     elif
choice == "3":
delete_task()     elif
choice == "4":
complete_task()
elif choice == "5":
      break
else:
      print("Invalid choice!")
```

OUTPUT:



Git Commands:

```
git init
git add .
git commit -m "Add original todo manager"

git branch feature/improvements
git switch feature/improvements

git add .
git commit -m "Add AI-assisted improvements"

git switch main
git merge feature/improvements

git log --oneline --all --graph
```

TASK3:

Prompt given to AI:

"Explain how to initialize a Git repository, create and switch to a new branch, add AI-assisted code modifications, commit changes with a meaningful message and merge the changes into the main branch. Provide the Git commands with explanations."

Old Code:

tasks = []

task = input("Enter task: ") tasks.append(task)

print("Tasks:", tasks)

task_no = int(input("Enter task number to delete: ")) tasks.pop(task_no

- 1)

```
print("Updated tasks:", tasks)
```

IMPROVED CODE:

```
tasks = []

def add_task():
    task = input("Enter task: ").strip()

    if task:
        tasks.append({"name": task, "completed": False})
print("Task added")     else:
        print("Task cannot be empty")

def view_tasks():     for i, task in
enumerate(tasks, start=1):         status =
"Completed" if task["completed"] else
"Pending"         print(i, task["name"], "-",
status)

def complete_task():
    view_tasks()

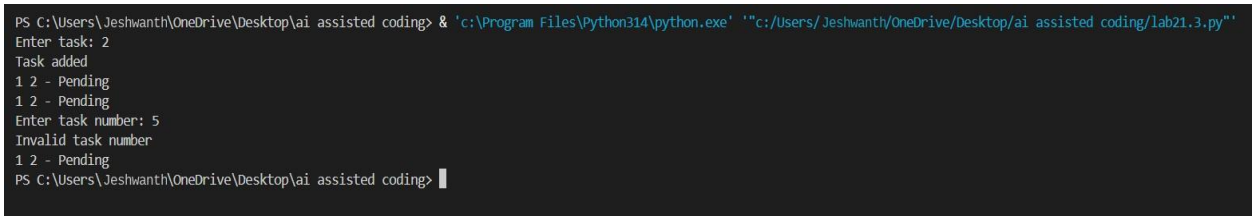
try:
        number = int(input("Enter task number: "))
```

```
        if 1 <= number <= len(tasks):
tasks[number - 1]["completed"] = True
print("Task completed")         else:
            print("Invalid task number")

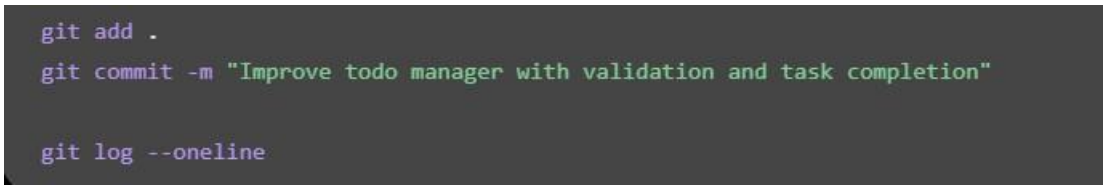

    except ValueError:
        print("Enter a valid number")


add_task() view_tasks()
complete_task()
view_tasks()
```

OUTPUT:



Git Commands:



TASK4:

Prompt given to AI:

"Create a simple Python Calculator program. Student A writes the basic calculator with addition, subtraction, multiplication and division. Student B uses AI assistance to improve the program by adding input validation, error handling and better code organization using functions. Explain how both students can collaborate using Git branches and merging."

Old Code:

```
num1 = int(input("Enter first number: ")) num2
= int(input("Enter second number: "))

print("1. Addition") print("2.
Subtraction") print("3.
Multiplication") print("4.
Division")

choice = int(input("Enter choice: "))

if choice == 1:
   print("Result:", num1 + num2) elif
choice == 2:
   print("Result:", num1 - num2)
elif choice == 3:
   print("Result:", num1 * num2) elif
choice == 4:
   print("Result:", num1 / num2)
else:    print("Invalid choice")
```

IMPROVED CODE:

```
def calculator():
try:
      num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
```

```
        print("\n1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")

        choice = input("Enter choice: ")

        if choice == "1":
            result = num1 + num2
elif choice == "2":
            result = num1 - num2
elif choice == "3":
            result = num1 * num2
        elif choice == "4":
if num2 == 0:
                print("Cannot divide by zero")
return
            result = num1 / num2
else:
            print("Invalid choice")
return

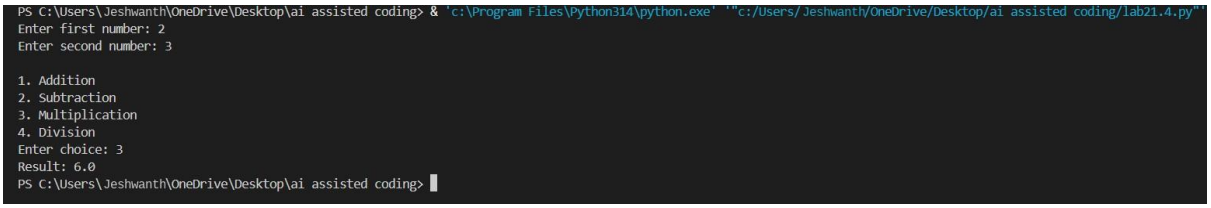
        print("Result:", result)

    except ValueError:
```

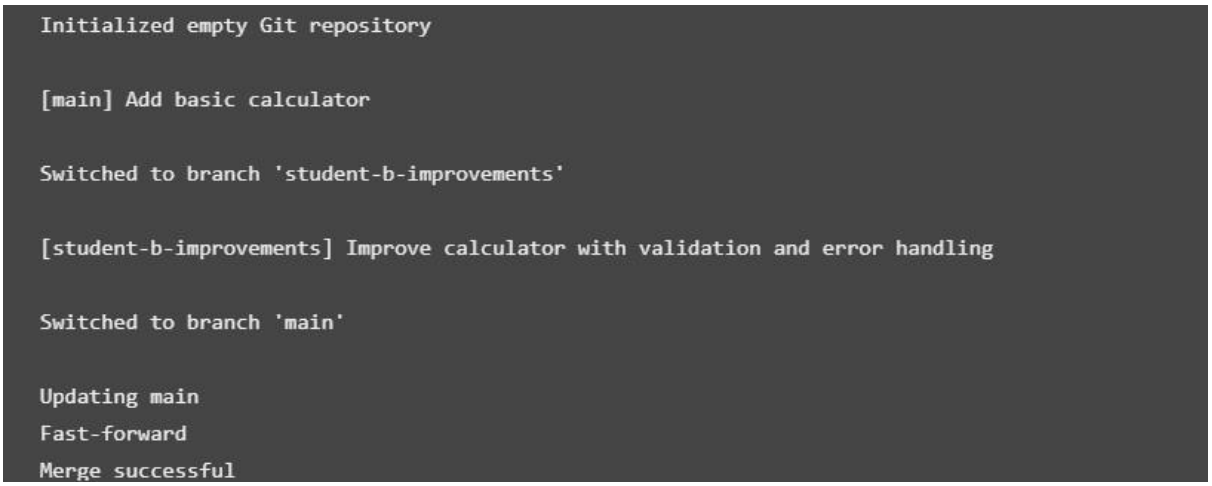
```
    print("Please enter valid numbers")


calculator()
```

OUTPUT:



Git Commands:



TASK5:

Prompt given to AI:

"Find and fix the syntax and logical errors in the following Python program. Explain the errors and provide the corrected code."

Old Code:

```
# Program to calculate the average of three numbers
```

```
num1 = input("Enter first number: ") num2
= input("Enter second number: ") num3 =
input("Enter third number: ")

total = num1 + num2 + num3 average
= total / 2

if average > 50
print("Pass") else:
print("Fail")
```

IMPROVED CODE:

```
# Program to calculate the average of three numbers

try:
   num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
num3 = float(input("Enter third number: "))

   total = num1 + num2 + num3
average = total / 3

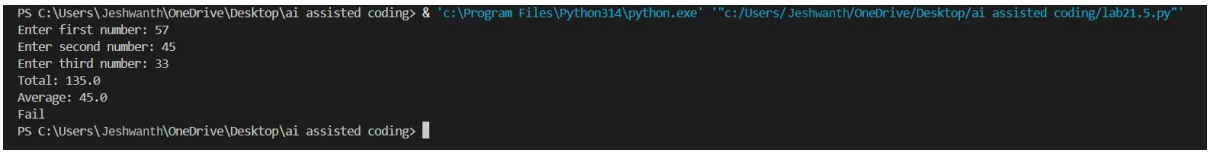
   print("Total:", total)
print("Average:", average)

   if average >= 50:
print("Pass")    else:
```

```
        print("Fail")

except ValueError:
    print("Please enter valid numbers")
```

OUTPUT:



COMPARISION:

Buggy Version	**Corrected Version**
Missing colon after if colon	Added the missing
Used input() without converting to numbers	Used float()
Calculated average by dividing by 2	Corrected division to 3
No invalid-input handling	Added try-except
Incorrect pass condition (> 50)	Uses >= 50

TASK6:

Prompt given to AI:

"Create a basic Python calculator with addition, subtraction, multiplication and division. Use AI to improve it by adding calculation history, input validation, division-by-zero handling and better code organization."

Old Code:

```
num1 = float(input("Enter first number: ")) num2

= float(input("Enter second number: "))


print("1. Addition")
print("2. Subtraction") print("3.

Multiplication") print("4.

Division")


choice = int(input("Enter choice: "))


if choice == 1:

   print("Result:", num1 + num2) elif

choice == 2:

   print("Result:", num1 - num2) elif

choice == 3:

   print("Result:", num1 * num2) elif

choice == 4:

   print("Result:", num1 / num2) else:

   print("Invalid choice")
```

IMPROVED CODE:

```
history = []
```

```
def calculator():
try:
    num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

    print("\n1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")

    choice = input("Enter choice: ")

    if choice == "1":
result = num1 + num2
operator = "+"      elif
choice == "2":          result =
num1 - num2
operator = "-"      elif
choice == "3":          result =
num1 * num2
operator = "*"      elif
choice == "4":          if
num2 == 0:
            print("Cannot divide by zero")
return
```

```
            result = num1 / num2
operator = "/"          else:
            print("Invalid choice")
return


        print("Result:", result)
        history.append(f"{num1} {operator} {num2} = {result}")      except
ValueError:
        print("Please enter valid numbers")

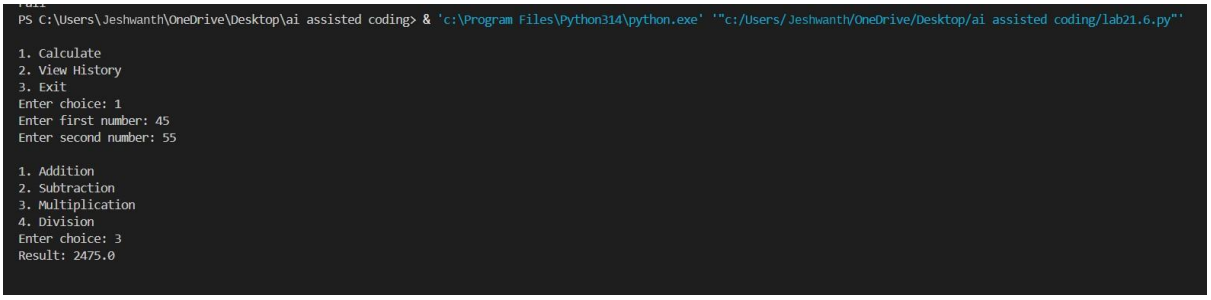

def show_history():
if not history:
        print("No calculation history")
else:
        print("\nCalculation History:")
for item in history:
print(item)


while True:
    print("\n1. Calculate")
print("2. View History")
print("3. Exit")


    choice = input("Enter choice: ")
```

```
    if choice == "1":
calculator()     elif
choice == "2":
show_history()
elif choice == "3":
        print("Exiting calculator")
break
    else:
        print("Invalid choice")
```

OUTPUT:



Analysis of Improvements

Basic Calculator	**AI-Enhanced Calculator**
Performs basic calculations	Performs calculations and stores history
No input validation	Handles invalid inputs
No division-by-zero protection	Prevents division by zero
No history feature	Displays previous calculations
Limited functionality	More user-friendly and reliable